Cover Page

Hack for Social Cause

• **Problem Statement/Concept Title**

Samagna · BanaSetu (ସମଗ୍ନ)

Minor Forest Produce Custody Ledger, Tare Verification & Real-Time Transit Shrinkage Audit Engine

• **Theme**

Rural development & livelihoods

Aligned with Government of Odisha 100% State-Funded LABHA Scheme & TDCCOL Framework

• **Team Name (Registered on portal)**

PRODERS

• **Team ID(Registered on portal)**

HSC|OR|00037

• **State/UT**

Odisha

Ravenshaw University, Cuttack

Live Deployment & Highlights

🌐 VERIFIED WORKING PROTOTYPE

https://samagna.netlify.app

1.4 Million Tribal Collectors
Similipal Biosphere Reserve & Mayurbhanj rural forest fringe.

100% Offline-First (Dexie DB)
Zero-signal edge ledger with automatic cloud sync upon signal recovery.

<300ms Shrinkage Engine
Separates allowable natural moisture loss from illicit transit pilferage.

Low-Literacy Odia Web Speech
Zero-typing touch interface + native Odia voice readout ("Listen to Rate").

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Problem – Proposed Solution

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Core problem/Proposed Solution (Describe your Concept/Idea/Solution/Prototype)

• Detailed explanation of the proposed solution

▪ **Offline-First Dual Ledger:**
Runs 100% offline at forest edge via Dexie IndexedDB. When connectivity resumes, batches sync idempotently to central PostgreSQL.

▪ **5-Tier Unified Custody Chain:**
Connects Primary Collectors, SHG Mahila Mandals, Aggregation Scale Managers, Depot Operators, and District Admins.

▪ **Calibrated Digital Scale Math:**
Integrates electronic scales with automatic tare subtraction (Gross - Tare = Net) to eliminate manual paper pen manipulation.

▪ **Dual Immutable Artifacts:**
Issues A6 durable waterproof bag stencil QR tags + bilingual Odia/English thermal physical receipts.

▪ **Live Prototype URL:**
https://samagna.netlify.app

• How does it address the problem

▪ **Dual-Ledger Exploitation Solved:**
Historically, village paper slips contradicted warehouse scales. BanaSetu binds all checkpoints to an immutable SHA-256 digital record.

▪ **300ms Shrinkage Engine:**
Natural moisture evaporation in 130km transit is calculated against botanical thresholds (default 1.5% absorbed by state LABHA fund).

▪ **Non-Accusatory Fair Trade Guardrail:**
Central depot transit discrepancies NEVER retroactively alter or reduce the primary gatherer's certified field payment.

▪ **Administrative Quarantine:**
Excess deficits (>1.5%) are quarantined to Bay 2 for supervisor review without holding up gatherer DBT payouts.

• Innovation and uniqueness of the solution

✦ **Low-Literacy Tribal UX:**
Zero-typing workflows, native Odia typography (Noto Sans Oriya), authentic botanical photography, and large touch targets.

✦ **Web Speech Odia Voice Rate Card:**
"Listen to Rate" audio button speaks official MSP rates aloud in Odia dialect for non-literate forest dwellers.

✦ **Scientific Moisture Segregation:**
First governance ledger recognizing botanical evaporation physics rather than accusing tribal collectors of shrinkage fraud.

✦ **Public Provenance Ledger:**
Public verifier (/verify/[code]) lets citizens and buyers inspect unbroken cryptographic custody from forest tree to bank DBT.

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Technical Approach

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• Technologies to be used (languages, frameworks, stack)

Client / Edge Next.js 14 (App Router), TypeScript, Tailwind CSS, Lucide Icons

Local Storage Dexie.js v4 (IndexedDB) for 100% offline local transactions & queue

Backend & Auth Node.js Edge Runtime, JOSE (HMAC-SHA256 JWT in HttpOnly cookies), CSP

Database PostgreSQL with Prisma ORM (15 relational models, migrations & audit log)

APIs & Audio Sliding-Window Rate Limiting, Web Speech Synthesis API (Odia Voice)

• Methodology and process for implementation (Process Flow)

1. **Village Collector / SHG:** Odia speech rate check ➔ single-tap produce pickup request
2. **Aggregation Scale Point:** Calibrated scale tare deduction ➔ A6 QR bag tag & thermal receipt
3. **Central Depot Reweigh:** 300ms Shrinkage Engine splits allowable moisture loss vs quarantine
4. **District HQ & DBT:** Route monitoring ➔ conflict resolution ➔ Public SHA-256 verification

✓ **Working Prototype Validated: 24 automated test suites (100% pass: i18n, a11y, RBAC, scale math).**

• Github/Gitlab link

https://github.com/Rudra4451/Samagna

Full production codebase including Next.js 14 App Router, Prisma ORM schemas, Dexie offline sync engine, and 24 verification test suites.

• Demo Video

📼 **Video Showcase Link:**
https://drive.google.com/file/d/1847Fmx0qJsOU9-rSgFcjvAF8E4dCnnsX/view?usp=sharing

🌐 **Live Interactive Prototype:**
https://samagna.netlify.app

• Extent of usage of AI tools

✦ **Web Speech Synthesis:** Regional Odia voice narration for non-literate collectors.

✦ **Taxonomy Alignment:** LLM-assisted botanical terminology for 17 forest species.

✦ **Automated Test Fuzzing:** AI-generated boundary cases for moisture shrinkage math.

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Impact Potential

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• Potential impact on the target audience

▪ **1.4 Million Tribal Primary Gatherers:**
Directly protects Santhal, Ho, and Kolha forest collectors across Mayurbhanj district and Similipal Biosphere Reserve.

▪ **Eradicating 8%–14% Income Leakage:**
Eliminates arbitrary middleman tare and transit weight cuts, preserving ₹2,400 to ₹4,800 annually per tribal household.

▪ **Zero Delayed Settlements:**
Replaces 30-day paper settlement delays with instant verified digital receipts and direct state treasury transfers.

▪ **Eradication of Middleman Exploitation:**
Audio rate cards in Odia ensure forest dwellers know statutory MSP rates before parting with harvested produce.

• Benefits of the solution (social, economic, etc.)

▪ **Economic (LABHA Scheme Compliance):**
Guarantees 100% adherence to Odisha statutory minimum support prices with calibrated digital scale math.

▪ **Social & Women SHG Empowerment:**
Local Mahila Mandals lead village-level aggregation, earning verified commissions and fostering community leadership.

▪ **Financial Inclusion (DBT Pipelines):**
Transitions unbanked tribal forest collectors into formal institutional banking through automated DBT disbursement files.

▪ **Ecological & Forest Governance:**
Tamper-evident QR bag stencils stop illegal harvesting and timber/produce smuggling across biosphere boundaries.

• Stakeholders

✦ **Primary Tribal Gatherers:**
1.4 million forest dwellers collecting Mahua flowers, Sal seeds, Wild Honey, Kendu leaves, and Kusum.

✦ **Women Self-Help Groups (SHGs):**
Village-level community aggregators (e.g. Badagaon Mahila Mandal) organizing collective dispatch and elder assistance.

✦ **Scale & Warehouse Operators:**
Village aggregation scale operators and Baripada central depot managers managing transit logistics.

✦ **State Administration & TDCCOL:**
Mayurbhanj District Administration, ST & SC Development Department, and state welfare authorities governing statutory MSP circulars.

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Feasibility and Risk Analysis

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• Analysis of the feasibility of the solution

▪ Zero Hardware Capex:
Runs as a responsive Progressive Web App (PWA) on low-cost Android smartphones and tablets already distributed to SHGs and village secretaries.

▪ 100% Offline Feasibility:
Client-side Dexie IndexedDB enables full offline functionality for weighment, receipt generation, and bag tagging in remote forest valleys.

▪ Policy Alignment:
Fully complies with Odisha Government's 100% state-funded LABHA Yojana operational guidelines and TDCCOL procurement schedules.

• Potential challenges and risks

⚠ Zero Cellular Connectivity:
Deep Similipal forest villages can be disconnected from network towers for weeks.

⚠ Digital Illiteracy Gap:
Elder tribal gatherers unfamiliar with smartphone screens, English menus, and digital inputs.

⚠ Rugged Field Conditions:
Rain, mud, humidity, and physical device drops during monsoon forest collection.

⚠ Scale Miscalibration / Fraud:
Mechanical drift or intentional tampering of electronic scales at village aggregation points.

• Strategies for overcoming these challenges

✓ Durable Offline Outbox:
Transactions commit to local IndexedDB with cryptographic ULID idempotency; auto-syncs when signal is restored.

✓ Odia Web Speech & Zero-Typing UX:
"Listen to Rate" audio readouts in native Odia + authentic crop photos eliminate typing completely.

✓ Dual Physical-Digital Artifacts:
A6 waterproof QR stencil tags on sacks + bilingual thermal receipts guarantee durable custody even if devices break.

✓ Daily Calibration & Tare Protocols:
System enforces daily zero-point tare calibration with digital receipt logs before batch sealing.

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Team

Team Composition (Team PRODERS · Ravenshaw University, Cuttack · Team ID: HSC|OR|00037)

• Roles

Rudra Pratap Swain
TEAM LEAD · UI/UX & SYSTEM ARCHITECT
Overall system architecture, user interface design system (editorial linework, Noto Sans Oriya, high contrast), dual-database offline sync protocol, Prisma ORM schemas, and JWT/RBAC security.

Prince Kumar Meher
MEMBER 1 · FRONTEND & RESEARCH
Next.js 14 client interfaces, Web Speech API Odia rate narration, digital scale tare calculators, and field research on Odisha LABHA scheme operational guidelines.

Dikshyant Ranjan Behera
MEMBER 2 · DEPLOY & TESTING
Cloud & container deployment pipelines, automated test harness execution (24 test suites: i18n, a11y, scale math, transit shrinkage), and continuous integration.

• Challenges

▪ **Asynchronous Offline Synchronization:**
Designing an idempotent queue that reconciles field weighments with zero duplicate records and zero race conditions.

▪ **Scientific Moisture Modeling:**
Deriving botanical moisture loss curves for 17 forest species during 130km transit to differentiate natural evaporation from transit pilferage.

▪ **Low-Literacy Interface Design:**
Crafting a zero-typing, audio-guided Odia UI that tribal gatherers can use with zero barrier.

• Any other that the team would like to share about their journey.

✦ **Born From Lived Reality:**
Originating from Odisha (Cuttack, Kalahandi, Sundargarh), our team personally witnessed tribal gatherers losing seasonal earnings to arbitrary shrinkage cuts and handwritten paper chits.

✦ **Built for Real-World Deployment:**
We engineered BanaSetu not as a theoretical presentation, but as a fully tested, production-grade social infrastructure with 24 automated test suites passing.

✦ **Our Guiding Mission:**
"Forest. People. Prosperity." — Ensuring economic dignity and transparent governance for every tribal forest steward.

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References

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- **Details / Links of the reference and research work**

1. Government Schemes & Institutional Policy Frameworks

- **Tribal Development Cooperative Corporation of Odisha Ltd. (TDCCOL)** — *Minor Forest Produce Procurement Circulars, Seasonal Quality Standards & Minimum Support Price (MSP) Schedules (2024).*
- **Scheduled Tribes & Scheduled Castes Development Department, Govt. of Odisha** — *Operational Guidelines for 100% State-Funded LABHA Scheme (Laghu Bana Jatiya Dradya Kraya).*
- **Ministry of Tribal Affairs (MoTA), Government of India** — *Mechanism for Marketing of Minor Forest Produce (MFP) through Minimum Support Price (MSP) & Development of Value Chain.*

2. Technical & Academic Research Standards

- **Kleppmann, M., Wiggins, A., van Hardenberg, P., & Tratt, L. (2019)** — "*Local-First Software: You own your data, in spite of the cloud*", ACM Onward! Essay Series.
- **World Wide Web Consortium (W3C)** — *Web Content Accessibility Guidelines (WCAG) 2.1 AA Specification & Speech Synthesis Interaction Standards for Low-Literacy Populations.*
- **Internet Engineering Task Force (IETF) RFC 7519** — *JSON Web Token (JWT) Architecture Standard & Universally Unique Lexicographically Sortable Identifiers (ULID).*

3. Live Codebase & Public Ledger Verification Artifacts

- **Source Repository:** `https://github.com/Rudra4451/Samagna` (Next.js 14, Prisma ORM, Dexie IndexedDB, 24 Automated Test Suites).
- **Live Web Prototype:** `https://samagna.netlify.app` | **Demo Video:** `https://drive.google.com/file/d/1847Fmx0qJsOU9-rSgFcjvAF8E4dCnnsX/view?usp=sharing`
- **Public Provenance Verifier:** Live verifiable endpoint (`/verify/[code]`) verifying unbroken SHA-256 custody chain from tree harvest to state bank DBT.